

4.3.9.30. TemperatureSetpointHoldDuration Attribute

This attribute SHALL indicate the period in minutes for which a setpoint hold is active. Thermostats that support hold for a specified duration SHOULD implement this attribute. The null value indicates the field is unused. All other values are reserved.

If this attribute is updated to a non-null value and the [TemperatureSetpointHold](#) is set to [SetpointHoldOn](#) and the [SetpointHoldExpiryTimestamp](#) is supported, the server SHALL update [SetpointHoldExpiryTimestamp](#) with a value of current UTC timestamp, in seconds, plus the new value of this attribute multiplied by 60.

If this attribute is set to null and the [SetpointHoldExpiryTimestamp](#) is supported, the server SHALL set the [SetpointHoldExpiryTimestamp](#) to null.

4.3.9.31. ThermostatProgrammingOperationMode Attribute

This attribute SHALL indicate the operational state of the thermostat's programming. The thermostat SHALL modify its programming operation when this attribute is modified by a client and update this attribute when its programming operation is modified locally by a user. The thermostat MAY support more than one active [ProgrammingOperationModeBitmap](#). For example, the thermostat MAY operate simultaneously in Schedule Programming Mode and Recovery Mode.

Thermostats which contain a schedule MAY use this attribute to control how that schedule is used, even if they do not support the [ScheduleConfiguration](#) feature.

When [ScheduleActive](#) is not set, the setpoint is altered only by manual up/down changes at the thermostat or remotely, not by internal schedule programming.

NOTE

Modifying the [ScheduleActive](#) bit does not clear or delete previous weekly schedule programming configurations.

4.3.9.32. ThermostatRunningState Attribute

This attribute SHALL indicate the current relay state of the heat, cool, and fan relays.

Unimplemented outputs SHALL be treated as if they were Off.

4.3.9.33. SetpointChangeSource Attribute

This attribute SHALL indicate the source of the current active [OccupiedCoolingSetpoint](#) or [OccupiedHeatingSetpoint](#) (i.e., who or what determined the current setpoint).

This attribute enables service providers to determine whether changes to setpoints were initiated due to occupant comfort, scheduled programming or some other source (e.g., electric utility or other service provider). Because automation services MAY initiate frequent setpoint changes, this attribute clearly differentiates the source of setpoint changes made at the thermostat.

4.3.9.34. SetpointChangeAmount Attribute

This attribute SHALL indicate the delta between the current active [OccupiedCoolingSetpoint](#) or [OccupiedHeatingSetpoint](#) and the previous active setpoint. This attribute is meant to accompany